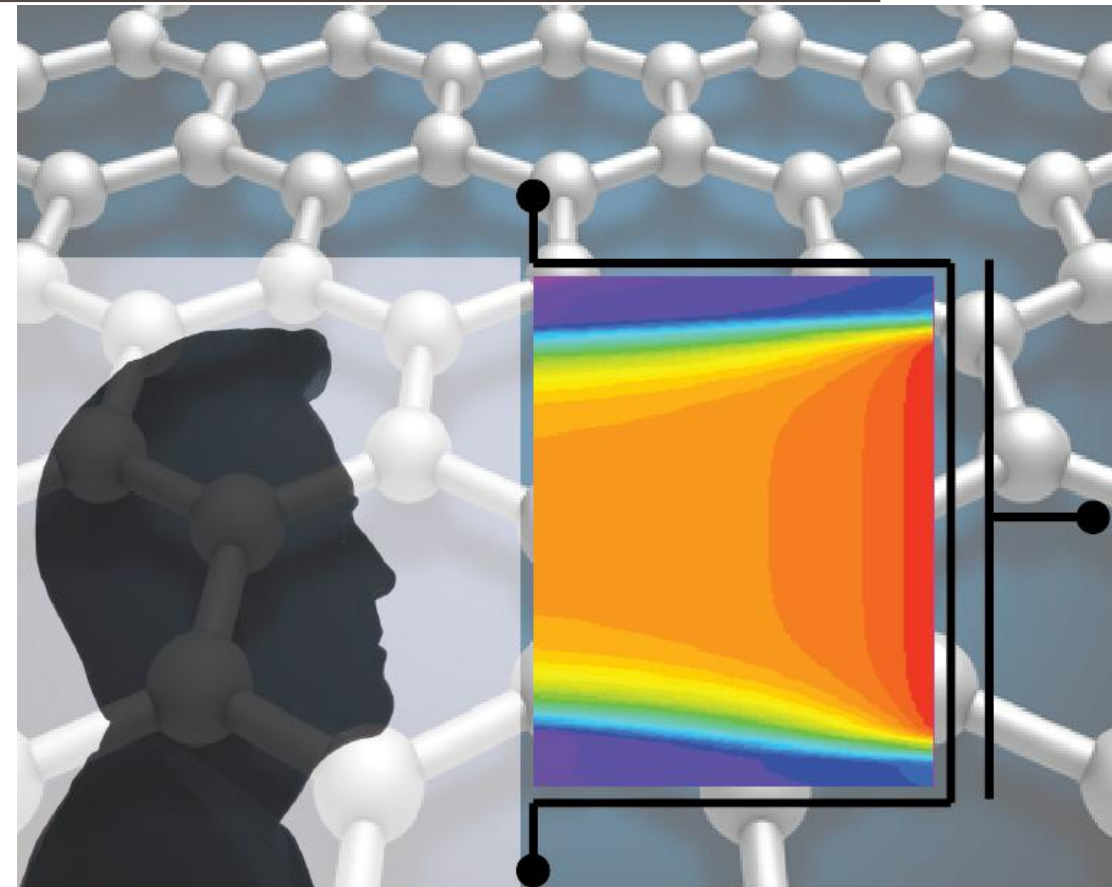


ADVANCED SOLID-STATE DEVICES

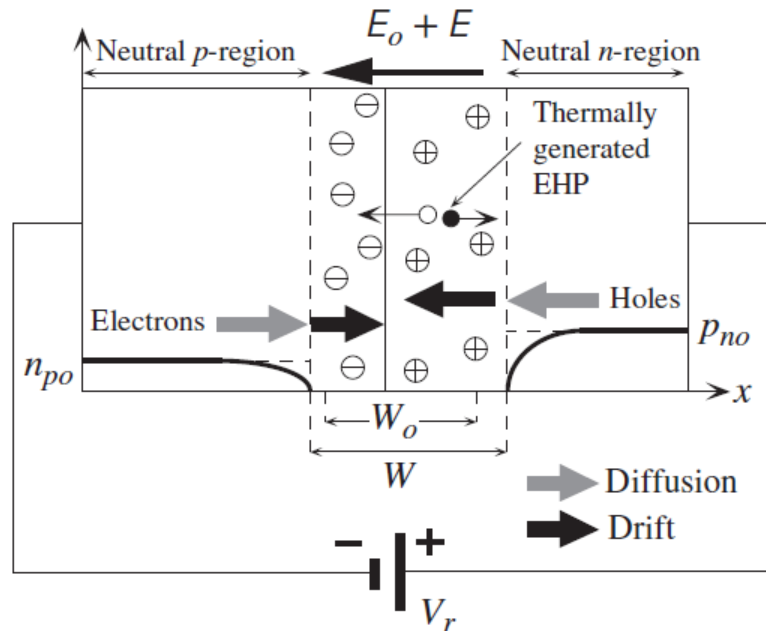
Week-11, Lecture-1

Sneh Saurabh
3rd April, 2025



Reverse-biased PN junction: Minority Carrier Profile

Minority carrier concentration



- Minority carriers (holes) on the n-side near the depletion region edge are swept by the electric field to the negative terminal
 - Continuous flow can be maintained by the diffusion of holes from the n-side bulk to the depletion region edge
- Minority carriers (electrons) on the p-side near the depletion region edge are swept by the electric field to the positive terminal
 - Continuous flow can be maintained by the diffusion of electrons from the p-side bulk to the depletion region edge
- Generation of electron-hole pairs in the depletion region will be another component of the reverse-biased PN junction current
 - Will enhance the total current

- Applied voltage drops mainly across the depletion region
- Electric field in the depletion region is larger than the built-in internal field

Bipolar Junction Transistor



Non-ideal Effects

Base Width Modulation (1)

- As reverse-biased voltage of BC junction is increased, the BC depletion region width increases and x_B decreases
 - Gradient of minority carriers in the base increases, and diffusion current increases
-
- This increase in collector current due to increase in reverse-biased BC junction is known as **base width modulation**
 - It is also called **Early effect**

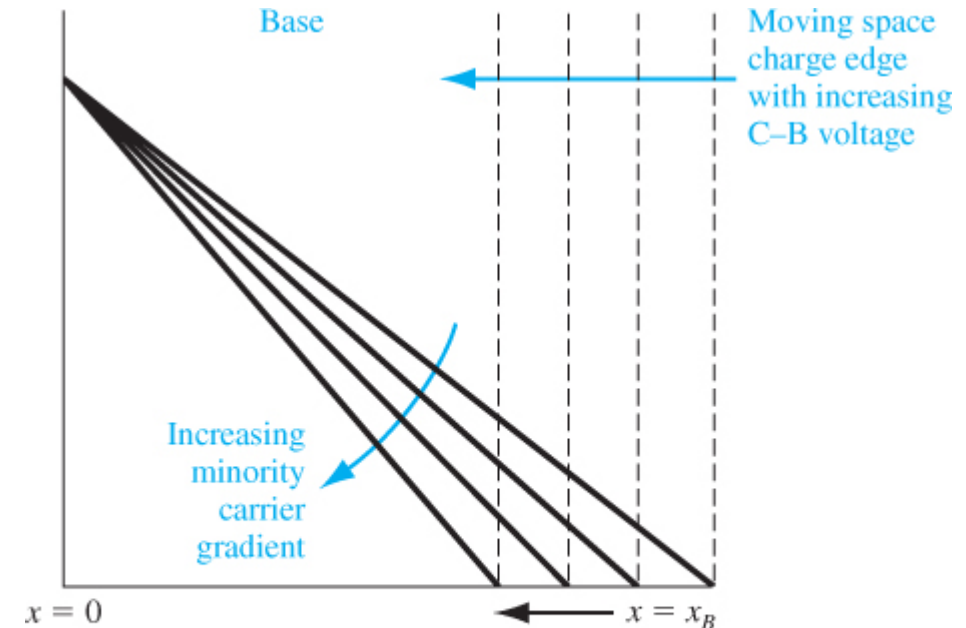


Figure 12.21 | The change in the base width and the change in the minority carrier gradient as the B–C space charge width changes.

Base Width Modulation (2)

- Ideally collector current is independent of V_{CE} when BC junction is reverse-biased (slope of curve zero)
- Due to early effect finite slope in the $I_C - V_{CE}$ curves
- If the curves are extrapolated to $I_C = 0$, they meet at a point V_A
- We refer to V_A as Early voltage (typically in the range of 100-300 V)

- We can define output resistance r_0 as follows:

$$\frac{1}{r_0} = \frac{dI_C}{dV_{CE}} = \frac{I_C}{V_A + V_{CE}}$$

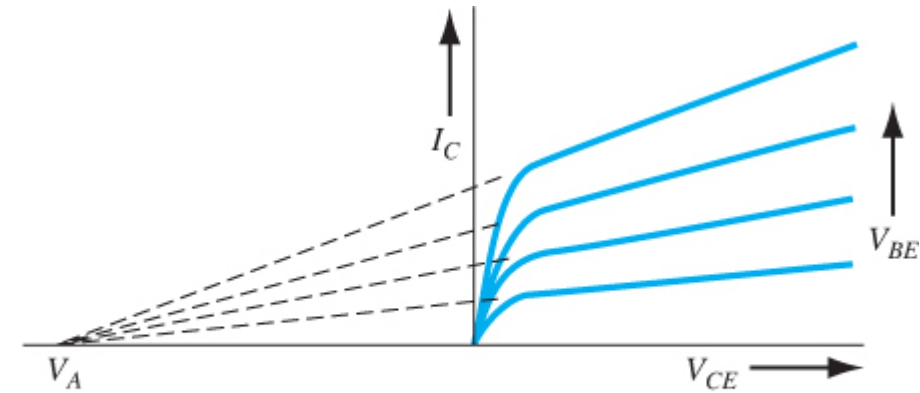


Figure 12.22 | The collector current versus collector–emitter voltage showing the Early effect and Early voltage.

- Variation in x_B can also be due to process-induced variations and lead to change in static characteristics

High-level Injection (1)

- When V_{BE} is high, injected minority can be more than majority carrier in the base
- Assuming quasi-neutrality, the majority carrier concentration in base at $x = 0$ will increase

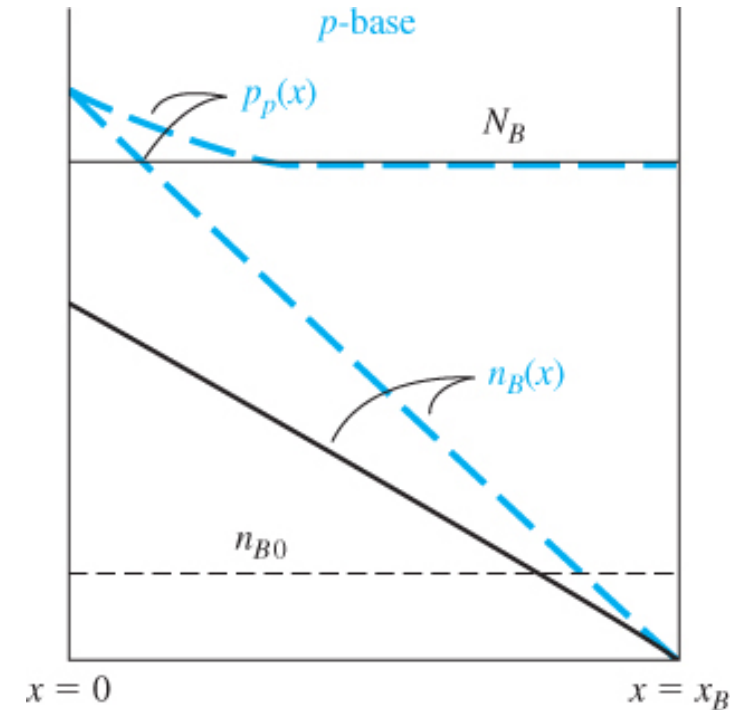


Figure 12.23 | Minority and majority carrier concentrations in the base under low and high injection (solid line: low injection; dashed line: high injection).

High-level Injection (2)

▪ Effect 1:

- The hole injection into emitter (due to diffusion) will increase
- Emitter injection efficiency reduces at high injection

▪ Effect 2:

- Low-level injection: $n_B = n_{B0} \left[\exp \left(\frac{V_{BE}}{kT/q} \right) \right]$
- High-level injection: $n_B = n_{B0} \left[\exp \left(\frac{V_{BE}}{2kT/q} \right) \right]$
- Collector current will rise at a slower rate with BE voltage

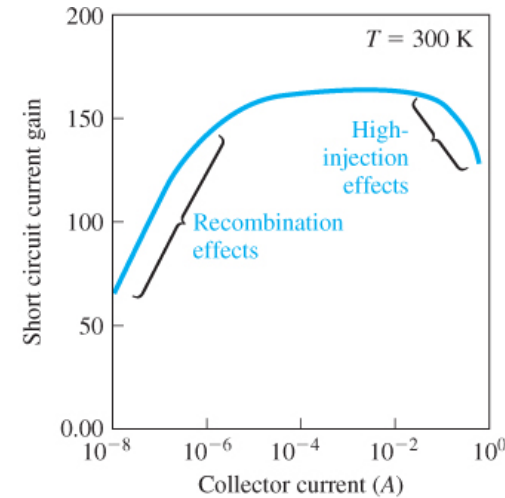


Figure 12.24 | Common-emitter current gain versus collector current. (From Shur [14].)

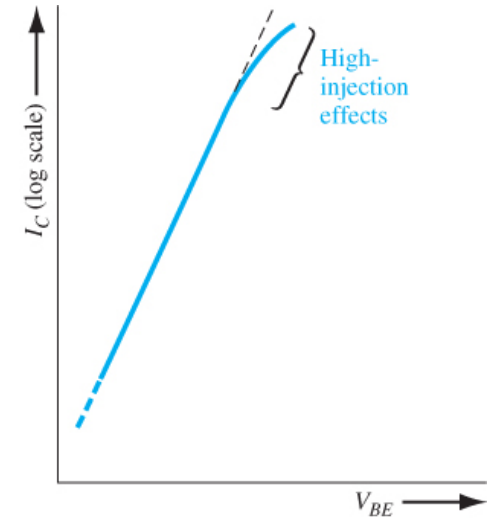


Figure 12.25 | Collector current versus base-emitter voltage showing high-injection effects.

Emitter Bandgap Narrowing

- When doping becomes very high, bandgap narrowing happens

- Intrinsic carrier concentration increases:

$$n_{iE}^2 = n_i^2 \exp\left(\frac{\Delta E_g}{kT}\right)$$

- Equilibrium minority carrier concentration in emitter

$$p_{E0} = \frac{n_{iE}^2}{N_E} = \frac{n_i^2}{N_E} \exp\left(\frac{\Delta E_g}{kT}\right) \text{ increases}$$

- Emitter injection efficiency starts decreasing at very high emitter doping

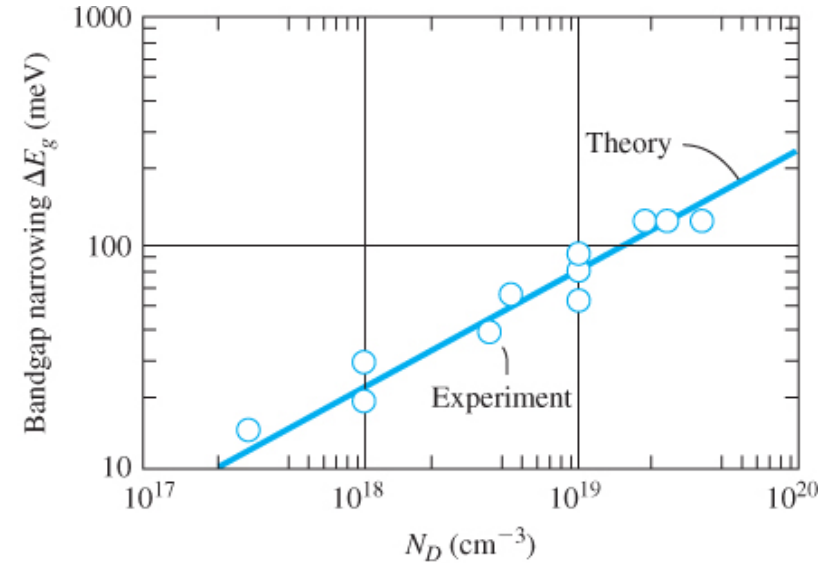


Figure 12.26 | Bandgap narrowing factor versus donor impurity concentration in silicon.
(From Sze [19].)

Current Crowding (1)

- Base has resistance, which lead to potential drop from the periphery to the interior
 - The voltage V_{BE} is higher in the outer edge of emitter and less in the center
-
- More electrons injected from the edge of the emitter than at the center
 - Current is crowded at the edge of the emitter

Effects of current crowding:

- Localized heating
- Localized high-injection effect

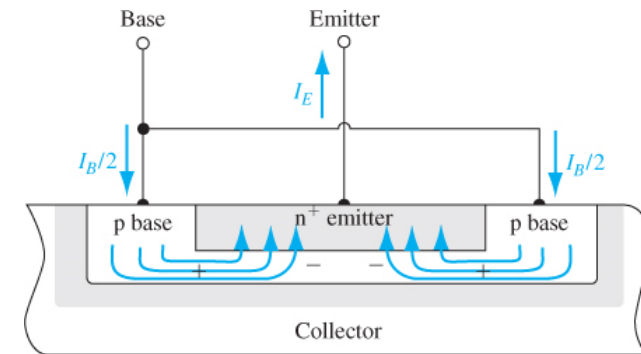


Figure 12.27 | Cross section of an npn bipolar transistor showing the base current distribution and the lateral potential drop in the base region.

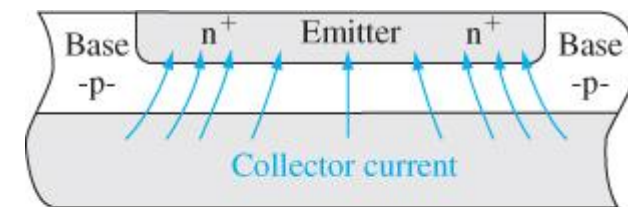


Figure 12.28 | Cross section of an npn bipolar transistor showing the emitter current crowding effect.

Current Crowding (2)

- For power BJT, we need to keep emitter area high
- To avoid current crowding narrow emitters (connected in parallel)
- Interdigitated design

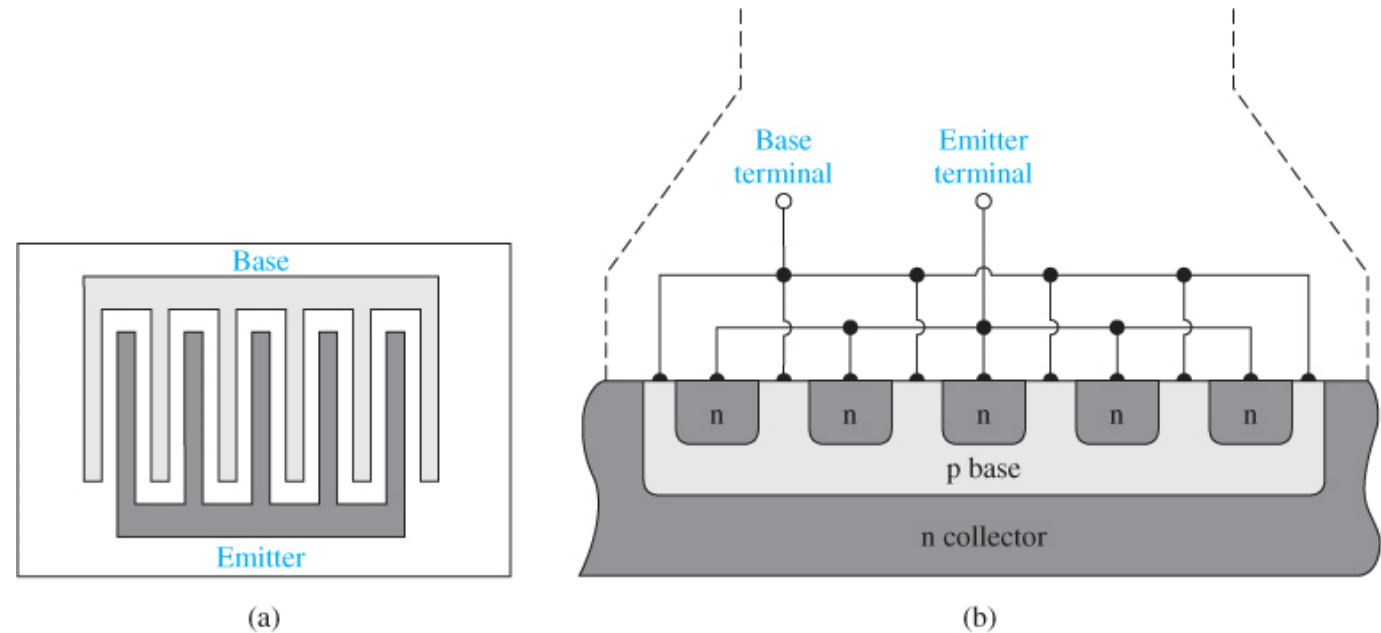


Figure 12.29 | (a) Top view and (b) cross section of an interdigitated npn bipolar transistor structure.

Breakdown Voltage

Two mechanisms of breakdown:

1. Punch-through
2. Avalanche breakdown

Punch-through:

- The reverse-biased BC depletion region meets BE depletion region
- Depletion region width on base side:

$$x_{dB} = \sqrt{\frac{2\epsilon_s(V_{bi} + V_{CB})}{q} \cdot \frac{N_C}{N_B} \cdot \frac{1}{N_C + N_B}}$$

- At punch-through $x_{dB} \approx x_{B0}$ [Assuming width of forward-biased BE junction is small]
- Assuming $V_{bi} \ll V_{CB}$:

$$x_{B0} = \sqrt{\frac{2\epsilon_s V_{pt}}{q} \cdot \frac{N_C}{N_B} \cdot \frac{1}{N_C + N_B}} \Rightarrow V_{pt} = \frac{q x_{B0}^2}{2\epsilon_s} \cdot \frac{N_B(N_C + N_B)}{N_C}$$

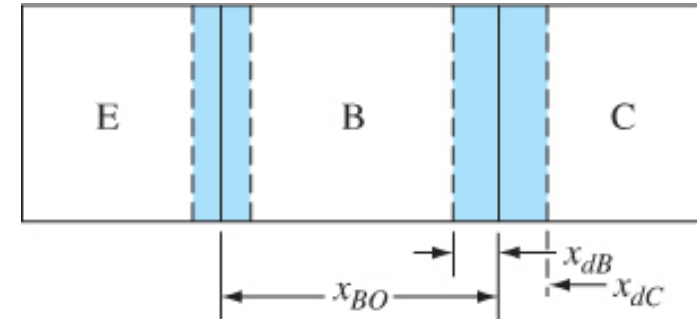


Figure 12.33 | Geometry of a bipolar transistor to calculate the punch-through voltage.

Avalanche breakdown:

- Similar to PN junction
 - Break down impacted by the current gain
-
- To increase breakdown voltage collector doping should be low

Bipolar Junction Transistor



Hybrid-Pi Model

Hybrid-Pi Model: Components (1)

- BJT normally used in linear amplifier circuits
- Small signal equivalent circuit important
- Used in forward-active mode

- Hybrid-Pi model built by considering BE, CE and BC PN junctions separately and then combining
- Consider idealized internal node E', B' and C'

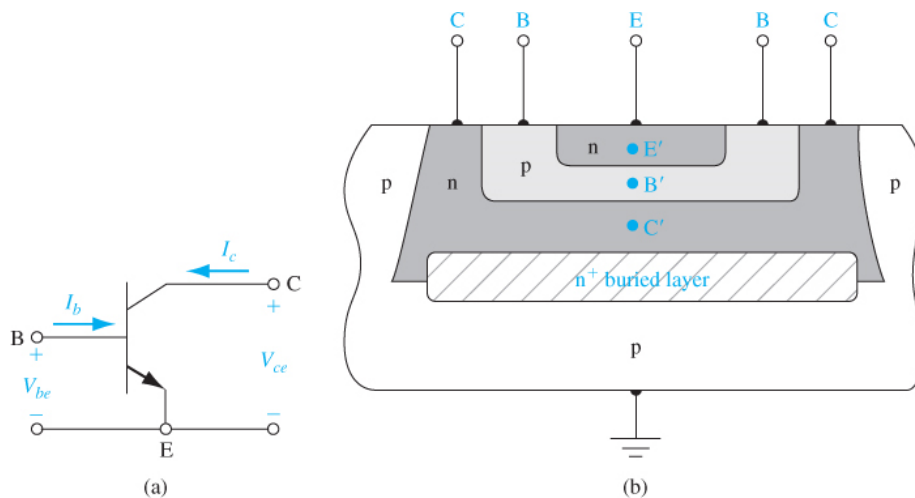
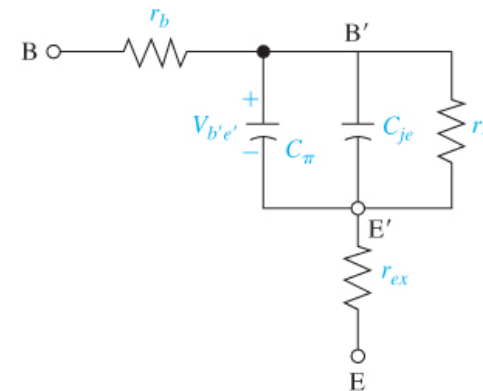


Figure 12.38 | (a) Common-emitter npn bipolar transistor with small-signal current and voltages. (b) Cross section of an npn bipolar transistor for the hybrid- π model.



BE junction (forward biased):

- r_b and r_{ex} series resistance
- r_π and C_π diffusion resistance/capacitance
- C_{je} junction capacitance

Hybrid-Pi Model: Components (2)

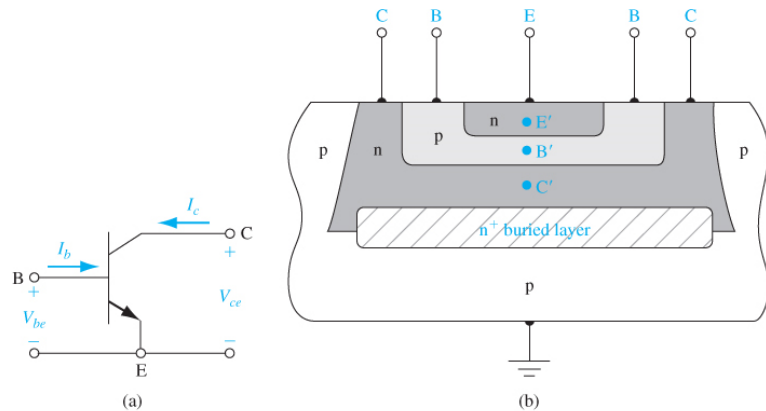
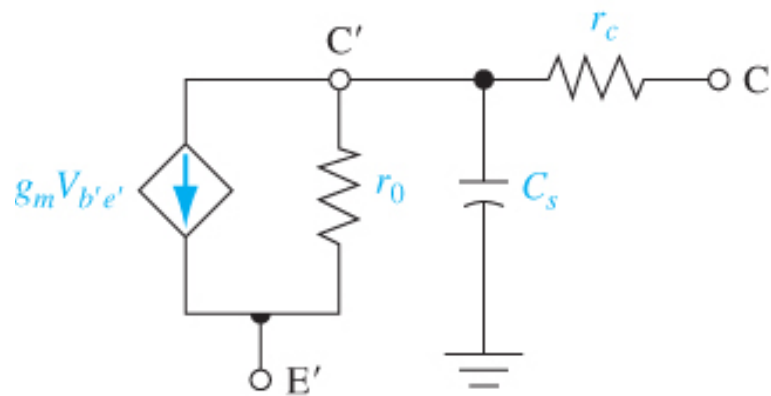


Figure 12.38 | (a) Common-emitter npn bipolar transistor with small-signal current and voltages. (b) Cross section of an npn bipolar transistor for the hybrid-pi model.

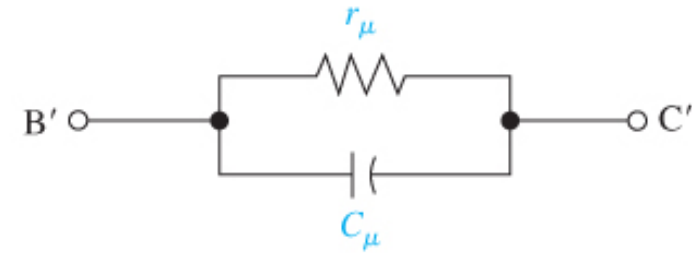


CE junction:

- $g_m V_{b'e'}$ current source, where

$$g_m = \frac{\partial I_C}{\partial V_{BE}} = \frac{I_C}{kT/q}$$

- r_c series resistance
- r_0 output resistance (due to Early effect)
- C_s collector to substrate capacitance



BC junction (reverse-biased):

- C_μ reverse-biased junction capacitance
- r_μ reverse-biased diffusion resistance (very high, can be neglected)

Hybrid-Pi Model: Equivalent Circuit

- Combine the models for terminal to get full hybrid-pi model
 - Can be used in computer simulation
-
- Can be simplified to get some intuitive understanding (though neglecting C_μ can lead to errors)

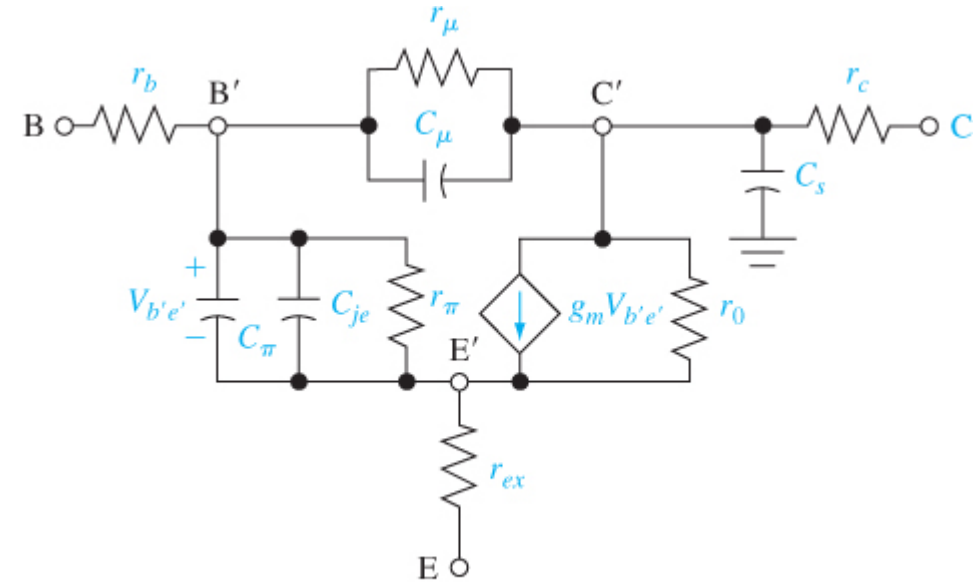


Figure 12.40 | Hybrid-pi equivalent circuit.

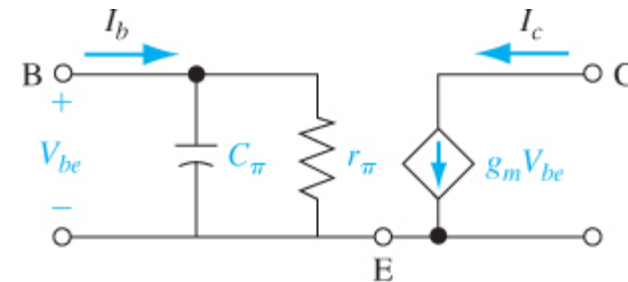


Figure 12.41 | Simplified hybrid-pi equivalent circuit.

Frequency Limitations

- Cutoff frequency f_T related to the total propagation delay τ_{EC} of carriers from emitter to collector:

$$f_T = \frac{1}{2\pi\tau_{EC}}$$

- τ_{EC} can be written as:

$$\tau_{EC} = \tau_E + \tau_T + \tau_D + \tau_C$$

τ_E = BE junction charging time (from hybrid-pi model)

$\tau_T = \frac{x_B^2}{D_B}$ Base transit time (depends on base width)

$\tau_D = \frac{x_{dbc}}{v_s}$ Transit time through the collector depletion region

τ_C = Collector capacitance charging time (from hybrid-pi model)

Device Simulation, Circuit Simulation, Device Models

Physics-based Models

- We studied physics of semiconductor devices (PN junction, Schottky junction, JFET, MOSFET, MESFET, BJT)
- We computed current using device physics
- The model that we developed are *physics-based model*

Approximations:

- We used approximations that are typically valid for devices of larger dimensions
- At smaller dimensions these models should be used with care (good to get intuitive understanding, not accurate for computing current, voltage, etc.)

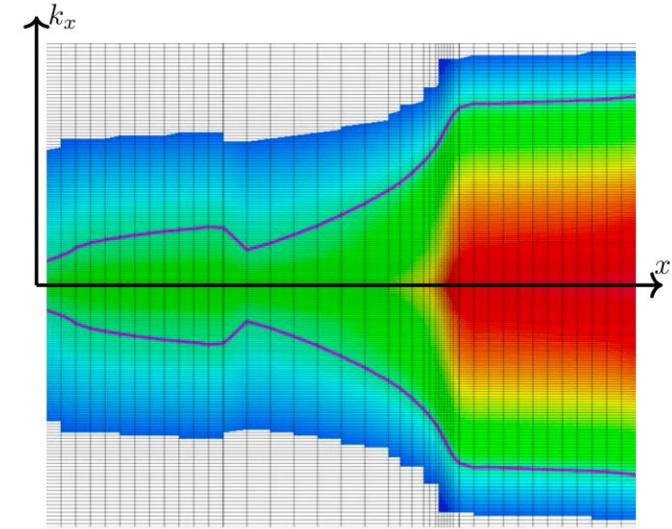
At smaller dimensions and for greater accuracy:

- We use *2D/3D device simulations* using tools that perform numerical computation

Device Simulation

- Divide the device into very-small sized grids and specify the grid in the tool
- If the grid size is sufficiently small, final result is independent of the grid size

- Tool internally formulate equations at each grid point with the help of:
 1. Poisson's equation
 2. Drift-diffusion equation
 3. Continuity equation
- The given voltage/bias acts as constraints for these equation
- Some reasonable value of material parameters are assumed



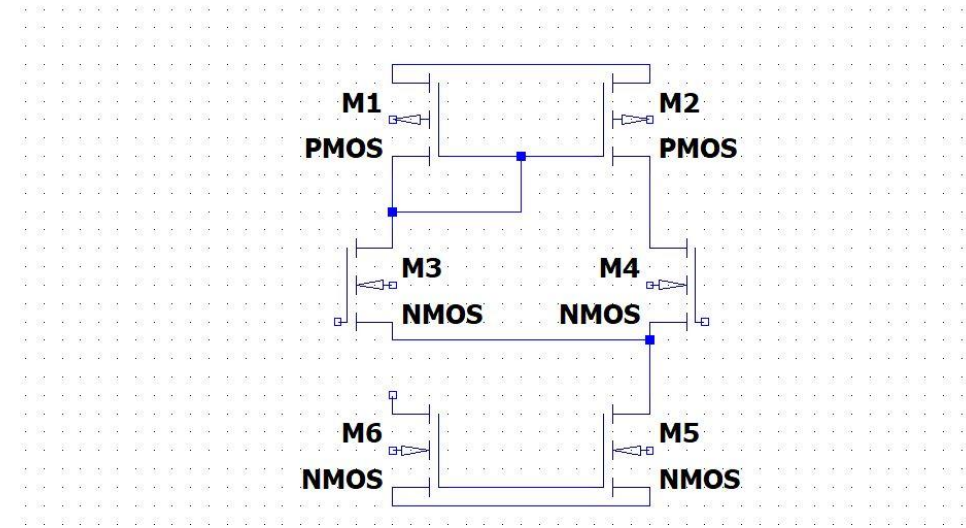
- These equations are represented as Matrix equation and solved using numerical techniques (such as NR/GS techniques)
- The results obtained are accurate
- Can be used for designing devices

Circuit Simulation

- We want to design circuits with the help of devices
- For designing circuits, we need to do circuit simulation: typically using SPICE/Virtuoso

- Device simulators are too slow to be plugged in with Circuit simulators

- For circuit simulation, we need to create abstract models of devices that can *quickly* give current for given bias and other parameters



- So, we need to develop device models for devices that allow circuit simulator to produce results quickly

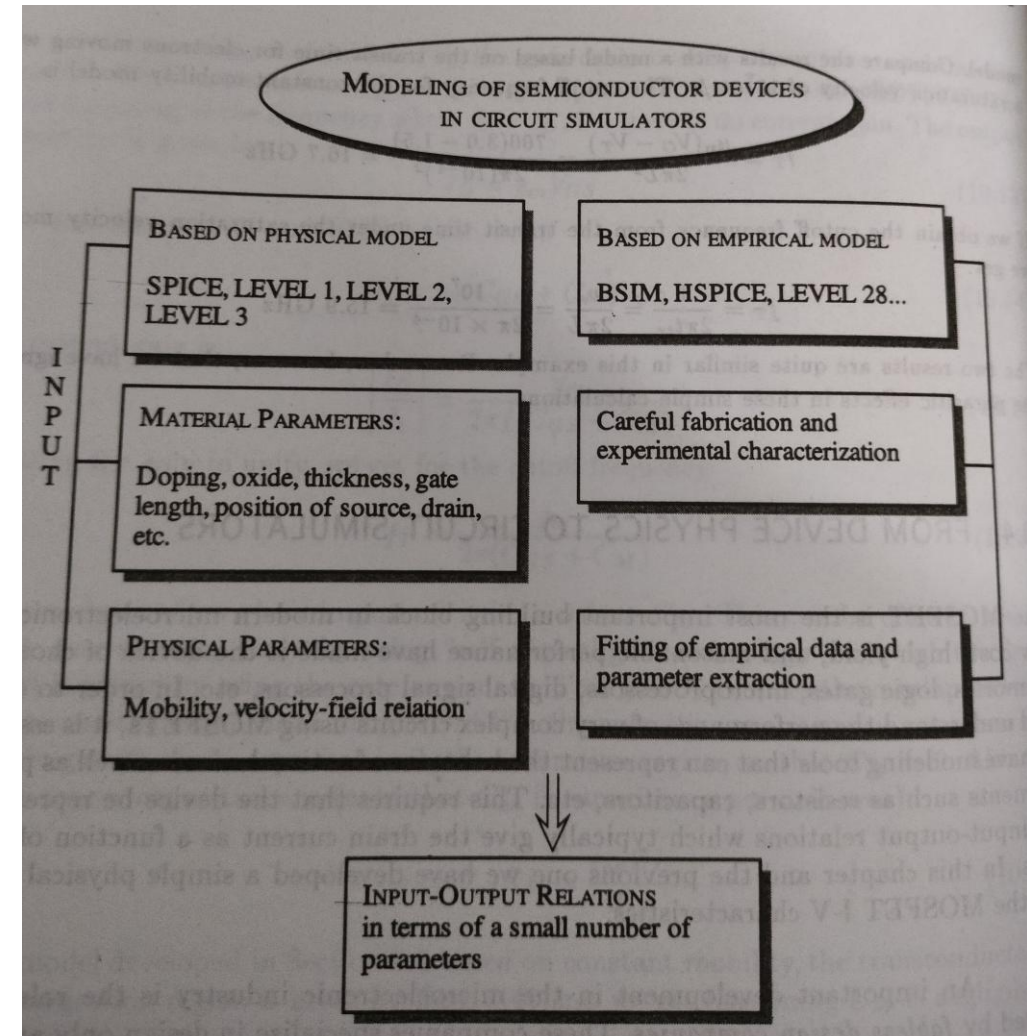
Device Models

- Earlier *physics-based models* were good enough for devices
- Now, we need to develop *empirical device models* for modern MOSFETs such as SOI MOSFET, FinFET etc.

Empirical device models: experimental data to fit equations with a number of adjustable parameters

MOSFETs: BSIM

BJT: Ebers-Moll Model, Gummel-Poon Model



References

- D. A. Neaman, "*Semiconductor Physics and Devices*"
- Kasap, Safa O. "*Electronic materials and Devices*", McGraw-Hill New York, 2006.